
Does an AI Have a Money Personality?

An Experiment in Machine Behavioral Economics

A teaching paper for undergraduate economics students

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Who this is for: a student who has taken intermediate microeconomics and has never studied artificial intelligence. You need to know what a utility function is, what an expected value is, and that some people are risk averse. You need to know nothing at all about how an AI model is built. Every AI idea in this paper gets a boxed explanation the first time it appears, and every economics idea gets a boxed refresher.

Abstract

Economists infer what people want from what they choose. This paper applies that method to language models, the systems behind AI assistants, and asks a question the field has mostly skipped: at which point in a model's construction does its handling of money and risk appear? We put 1,680 identical money questions to eleven versions of four AI systems, including four published snapshots taken at successive stages of one model's training. Five results follow. Freshly pre-trained models answer with total confidence and pick the larger of two sure payments barely more often than a coin flip, between 55.5 and 63.9 percent of the time, so their answers are not preferences in any usable sense. The measurable behavior appears at the first and simplest stage of assistant training, supervised fine-tuning, which accounts for about three fifths of the total change and, when the model must answer at once, is the only stage at which our estimated preferences move; the two later human-feedback stages keep pushing the behavior in the same direction without moving those estimates. The same finished model behaves like two different subjects depending on whether the question arrives as a chat message or as raw text to complete, and which format works reverses across model families. Rewording a bet so its downside reads as a loss changes gambling by -30.2 points in one system and $+62.6$ in another, so the famous framing effect is a property of a particular training recipe rather than of the technology. Finally, instructing a model in plain language to maximize expected value raises compliance from about 45 to 54 percent, close to indifference, while the same instruction is followed almost perfectly on questions requiring no arithmetic. Allowed to work the arithmetic out in writing before answering, the same models follow the rule 71 to 96 percent of the time, and the framing effect changes with the answering mode: one model's reaction to loss wording reverses direction, through a bookkeeping error it makes over and over. One added sentence removes that error, and paying the model through training does not teach it the arithmetic. We treat the interpretation of the second result as a lesson in identification: the data cannot separate training that created these dispositions from training that made a latent disposition readable, and we show why the policy conclusion holds under either story.

1. The question, and why an economist can answer it

Start with a question you can answer yourself before reading any further.

The cover question

You must pick one of two payment options.

Option A: receive \$86 for certain

Option B: a 43% chance to receive \$210, otherwise nothing

Pick the option you prefer.

Most people take a few seconds with that one. The gamble is worth more on average, and it leaves you with nothing more than half the time. Which option you pick says something about how you handle risk, and economists have been learning about people by asking questions of exactly this shape since the 1970s.

We asked AI models thousands of such questions. Then, for one model, we asked the same questions at four separate points in its construction and watched what changed. The headline of the paper is that almost everything about how this model handles money arrives at a single manufacturing step, and that step is the first and least glamorous one.

Before any of that can mean anything, two vocabularies have to be in place. The first is the economics, which you have seen. The second is the AI, which we assume you have not.

1.1 The economics you already have

KEY CONCEPT • Revealed preference

The founding move of choice theory. Rather than ask a person what they value, watch what they choose when the options are priced, and infer the preference that would rationalize the choices. The method needs no interview, no self-report, and no cooperation from the subject. It needs only that the subject makes choices and that the analyst controls the menu. That last condition is what makes the method portable to a machine: an AI model will not tell you its utility function, and would have no reliable access to it if asked, but it will pick A or B.

KEY CONCEPT • Expected value

The average payoff of a gamble, weighting each outcome by its probability. For a lottery paying h with probability p and nothing otherwise, $EV = p \cdot h$. In the cover question, $EV = 0.43 \times \$210 = \90.30 , against a sure \$86. The gamble is worth \$4.30 more on average. A decision-maker who cared only about the average would take the gamble.

KEY CONCEPT • Risk aversion

Most people do not care only about the average. Offered a fair coin flip between \$0 and \$100 against a sure \$50, the majority take the \$50, which means the certainty is worth something to them beyond the money. In the standard treatment this is curvature in the utility function: u is concave, so $u(\$50) > 0.5 u(\$0) + 0.5 u(\$100)$. A risk-averse person can decline the cover

gamble even though it pays more on average, and nothing is wrong with them for doing so.

WORKED EXAMPLE • Pricing the cover gamble, and why we need a second kind of question

Take the cover question apart. The certain option pays \$86. The gamble pays $0.43 \times \$210 = \90.30 on average. So:

- A **risk-neutral** agent, who maximizes expected value and nothing else, takes the gamble. \$90.30 beats \$86.
- A **risk-averse** agent may decline. The gamble's \$4.30 premium may not compensate for the 57 percent chance of walking away with nothing.

Both answers are defensible. That is the problem. If a subject picks the sure \$86, we cannot tell whether we are watching a coherent risk-averse decision-maker or a subject who is not engaging with the question at all. A single answer to a single interesting question carries no information about whether the subject is paying attention.

So we ask a second kind of question, where only one answer is defensible:

Option A: receive \$142 for certain

Option B: receive \$179 for certain

No probability, no risk, no taste involved. \$179 is more money than \$142. Every coherent preference ordering picks B. We call these the **dominant controls**, or informally the free money questions, and there are 80 of them scattered through the instrument. They function exactly the way an attention check functions in a survey. A subject who fails them has not revealed unusual preferences. That subject has revealed that its answers are not about the money.

Hold onto that distinction. It carries the first finding of the paper and it is the cheapest methodological contribution the project makes.

1.2 The AI you probably do not have

KEY CONCEPT • What a language model is

A language model is a function that takes a stretch of text and returns a probability distribution over what comes next. Give it **The capital of France is**, and it returns a list assigning high probability to **Paris**, lower probability to **a**, lower still to **located**, and so on across a vocabulary of tens of thousands of pieces. That is the entire mechanism. Everything an AI assistant appears to do is that operation run repeatedly, each predicted piece appended to the text and fed back in. The model has hundreds of millions to hundreds of billions of adjustable numbers, called **weights**, and training is the process of setting them.

KEY CONCEPT • Pre-training

The first and by far the most expensive training stage. The model is shown an enormous quantity of text, a large fraction of the public internet plus books and code, and its weights are adjusted so that it predicts each next piece of text better. That is the whole objective. It is never told what a helpful answer looks like, never told to follow an instruction, and never rewarded for being correct. A model in this state is called a **base model**: it has read a huge chunk of the internet and nothing else. Asked a question, it does not answer. It continues the text, which is a different behavior that sometimes resembles answering.

KEY CONCEPT • Supervised fine-tuning

The first stage of turning a base model into an assistant, usually abbreviated SFT. Humans write or curate a set of example exchanges, each a request paired with a good response, and the model's weights are adjusted to make those responses more likely. The mechanism is identical to pre-training, next-piece prediction, and only the data has changed: from whatever was on the internet to a curated set of demonstrations of being helpful. This stage is comparatively cheap, comparatively unfashionable, and it is where our headline result lands.

KEY CONCEPT • Preference training

The stage that follows. Humans are shown two candidate responses to the same request and asked which they prefer. Those comparisons are used to push the model toward the responses people picked. Two implementations dominate: reinforcement learning from human feedback (RLHF), which trains a separate scoring model and then optimizes against it, and direct preference optimization (DPO), which skips the scoring model and adjusts the weights directly from the comparisons. The distinction does not matter for this paper. What matters is that this is the stage where human judgment enters the weights, and it is therefore the usual suspect whenever a model displays a recognizably human quirk.

The ordering is what makes the study possible. Pre-training, then supervised fine-tuning, then preference training, then a final polish, and the finished assistant is what the public uses. Nearly all published work on AI decision-making tests only that finished assistant. Testing the finished product tells you the behavior exists. It cannot tell you which step installed it, which is the question an auditor, a procurement officer, or a regulator would actually want answered.

2. How we asked

2.1 The instrument

The full instrument is a grid of 1,680 questions. It is built from 40 gambles, generated procedurally so that none of them is a famous textbook problem the model might have memorized. Each gamble is rewritten five ways using different wordings, and each wording is shown in both possible orders so that a model with a bias toward the first option is caught by the design. Each gamble also appears under three framings, described below. The 80 free money questions sit inside the same grid.

Only one feature moves at a time. Any change in behavior between two cells traces to the one feature that differs between them, which is the entire reason for building an instrument this way rather than writing a few questions by hand.

KEY CONCEPT • Framing and reference dependence

Kahneman and Tversky's central observation is that people evaluate outcomes as gains and losses measured from a reference point, rather than as final states of wealth, and that losses loom larger than equivalent gains. The practical consequence is that the same choice, described two ways, gets two different answers. Standard theory says this should not happen. If two options leave you in identical final positions, the wording is decoration.

Our instrument tests exactly that. Here is the same gamble under two framings:

GAIN: Term A: a 64% chance to gain \$98 and a 36% chance to gain nothing
Term B: a certain gain of \$63

MIXED: You have been given \$63 up front.
Arrangement A: keep your \$63 with no change
Arrangement B: a 64% chance to rise to \$98 (a gain of \$35) and a
36% chance to drop to \$0 (a loss of \$63)

Final wealth is identical in both. Only the reference point moves, from \$0 to \$63, which turns part of the gamble's downside into a described loss. The **frame gap** is the difference in how often the model gambles between the two: mixed minus gain, in percentage points. A frame-invariant decision-maker scores zero.

2.2 Reading the answer

KEY CONCEPT • Why we read the model's internal odds instead of sampling its text

We never make the model type a reply. Recall that a language model returns a probability distribution over the next piece of text. At the position where the answer letter belongs, we read that distribution directly and record the probability it assigns to A against the probability it assigns to B, normalized to sum to one. That number is our observation: a probability of choosing the gamble, per question, not a binary choice.

Three advantages follow. First, precision. Sampling text 25 times to estimate a choice frequency gives a noisy estimate of the same quantity one forward pass gives exactly. Second, cost: one pass per question instead of 25. Third, and most important for this study, base models cannot be made to answer a question in a usable format, so any method requiring a parseable typed reply simply cannot be run on them. Reading the distribution works on every model at every training stage, which is what makes the stage-by-stage comparison possible at all.

There is also a statistical benefit. These readings are deterministic: the same question always returns the same numbers. All the randomness in the experiment is randomization we control, in the wordings, the orders, and the payoff grid, which is what our confidence intervals are computed over.

Since the observations are probabilities rather than yes-or-no choices, the estimation uses the

fractional-response likelihood of Papke and Wooldridge (1996) in place of the usual binary logit, with everything else in the structural model unchanged. That machinery is described in the technical companion to this paper and is not needed to follow the results here. The two quantities that carry the argument require no model at all:

- **Dominance compliance:** the average probability placed on the strictly larger sure amount, across the 80 free money questions. An engaged subject scores near 100. A coin flip scores 50.
- **The frame gap:** how much more often the model gambles under loss wording than under gain wording, on bets that are arithmetically identical. Frame invariance scores zero.

2.3 The subjects, and the one that makes the study work

KEY CONCEPT • Open-weight models, and why OLMo’s published snapshots matter

A **closed** model such as Claude or GPT is available only through a service. You send text, you get text back, and the weights stay with the developer. An **open-weight** model publishes the trained numbers themselves, so anyone can download the model and run it on their own hardware.

Open weights alone are not enough for this study. Most open-weight releases publish two things: a base model and a finished assistant. Comparing those two tells you that post-training changed the behavior, without telling you which of the three or four post-training stages did it.

OLMo-2, from the Allen Institute for AI, is the exception and is the reason this project exists in its current form. It publishes a snapshot of the weights after each stage along one training path: base, after supervised fine-tuning, after preference training, and the finished assistant. Four snapshots, one lineage, no branching. Asking identical questions of all four turns a coarse before-and-after comparison into a stage-level attribution. No frontier developer publishes anything comparable, which is a fact we return to in Section 9.

The full subject list is eleven runs: Qwen2.5-7B and Llama-3.1-8B, each as a base model and as a finished assistant, plus the four OLMo-2 stages, plus every finished assistant run a second time under an alternative question format. Claude Haiku, a closed frontier assistant measured in the earlier phases of this project, supplies a reference point. Seventeen runs in total including the instruction arms of Section 7, and 28,560 elicited choice probabilities.

3. Finding 1: A confident answer is not an answer

Figure 1 shows the free money check.

Every base model emits an answer letter with high confidence. The probability mass on A plus B has a median above 0.90 in every run, meaning the models are treating the prompt as an A-or-B question and committing to a letter. Then they fail the easiest question in the instrument. OLMo-2 takes the larger sure amount 55.5 percent of the time, just over half. Llama manages 62.3 percent and Qwen 63.9 percent, so roughly five times in eight. Claude Haiku, a finished assistant, scores 100.0 percent.

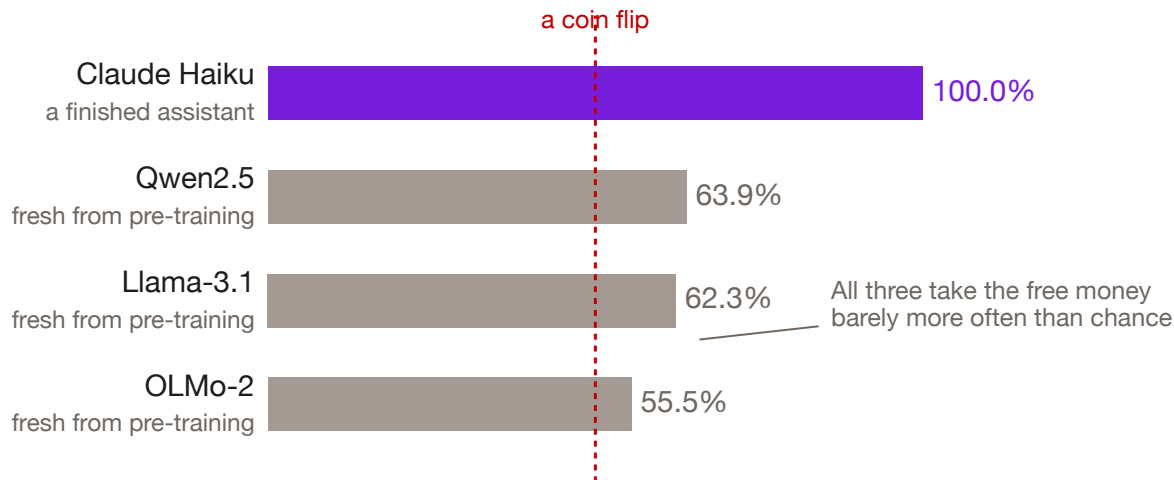


Figure 1: How often each model takes the strictly larger sure payment, with no gambling involved. The horizontal axis is the share of the 80 free money questions answered correctly, so the bars run from left (never) to right (always). The dotted red line at 50 percent is what pure guessing would produce. Read the three gray bars first: all three freshly pre-trained models sit within 14 points of a coin flip. The purple bar is a finished assistant, which never misses. Nothing about risk or taste is being measured here, only whether the subject is comparing two amounts of money at all.

This is the result that reframes everything after it. The project was designed to ask whether post-training bends a model’s existing preferences toward or away from human patterns. The base models show there is very little to bend. A subject that picks \$179 over \$142 barely more often than chance does not have unusual preferences. It has no measurable preferences, and any utility function estimated from its choices is describing a pattern closer to randomness than to taste. Wang, Li, and Chen (2025) found the same thing from the other direction: on the same Llama and Qwen base models, no utility function they tried fit the choices well. The free money check says why.

The generative story is worth stating plainly, because it is easy to misread the base model as strange rather than as absent. A base model has been trained to continue text plausibly. Shown four example questions with answer letters and then a fifth question, it produces a plausible fifth answer letter. Producing a plausible letter and comparing two amounts of money are different operations, and the base model is performing the first one.

WORKED EXAMPLE • Why the confidence is the trap

Suppose you skipped the free money questions and went straight to the interesting ones. OLMo-2 base would hand you 1,600 confident answers. You could fit a utility function to them. The optimizer would converge. You would get parameters, standard errors, and a plot. Every diagnostic short of the free money check would look normal.

Those parameters would describe nothing. This is why an evaluation of an AI system should report its engagement check the way a survey reports its response rate: the number is cheap, it is 80 questions, and without it a published result can be entirely an artifact.

4. Finding 2: Most of the change comes at supervised fine-tuning

Figure 2 is the core of the paper.

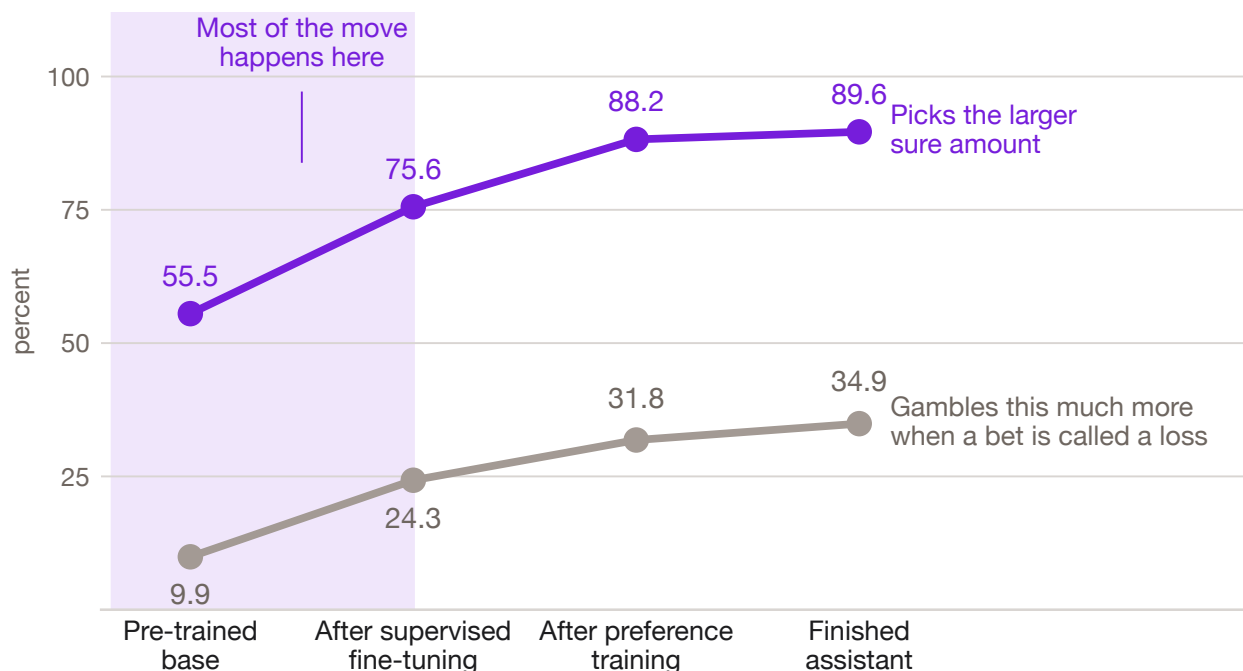


Figure 2: The same questions asked at four points in one model’s construction. The horizontal axis is the training timeline, running left to right from the pre-trained base to the finished assistant. The vertical axis is percent. The purple line is the free money check from Figure 1, and the gray line below it is the frame gap, the extra gambling the model does when a bet is reworded as a loss. Look at the shape rather than the levels: both lines make their largest move in the shaded band on the left, between the base model and supervised fine-tuning, then keep rising more slowly. Every checkpoint was asked in the identical format, so nothing here is a difference in how the question was posed.

Read the purple line first. The free money check moves from 55.5 to 75.6 percent at supervised fine-tuning, a jump of 20 points. Preference training adds another 12.6, to 88.2, and the final polish adds 1.4, to 89.6. The gray line has the same shape. The frame gap opens from 9.9 to 24.3 points at supervised fine-tuning, then rises to 31.8 and 34.9. Of the total change from base to finished assistant, supervised fine-tuning accounts for about three fifths on both lines, and preference training for most of the rest. Every step is larger than chance variation across the questions would produce, including the later ones.

The structural estimates tell a sharper story. Two of the base model’s three fitted parameters sit at the edge of their possible range. Its loss-aversion parameter is zero, which means it gives no weight at all to the losing side of a bet, and its curvature is at its maximum. After supervised fine-tuning the parameters move to interior values, and then they stop. The loss-aversion parameter moves by -0.06 with an interval of $[-0.23, +0.11]$ at the preference stage and by -0.02 with an interval of $[-0.18, +0.13]$ at the final stage. Every interval covers zero.

So the two kinds of evidence disagree after supervised fine-tuning, and the disagreement is informative. The raw behavior keeps moving; the fitted preferences do not. The later movement shows up instead as a worse fit: the model explains the preference-trained checkpoints less well than the supervised one. Preference training is changing something real that our model of preferences does not describe.

What that is remains open.

Stated as the paper’s central claim, in two parts: in this family, on these instruments, supervised fine-tuning is the largest single step and the only one at which the estimated preferences move; and the measurable profile is in place before preference training begins.

That ordering is a surprise. Preference training is where human judgment enters the weights, and it is the standard explanation offered when a model shows a human-like bias. Here it arrives to find the bias already present, and it amplifies what it finds.

One qualification belongs here, although its evidence comes in Section 7. Every number in this section comes from models answering at once, with a single letter. When the same checkpoints may reason in writing before answering, the later stages stop being quiet. The fine-tuned model still gambles 37 points more under loss wording, while the checkpoint right after preference training barely reacts, at about 1 point, and the finished assistant stays close to that. The central claim is therefore a claim about snap answers. How the model deliberates is set later, by preference training.

4.1 Did training create the preferences, or reveal them?

This is where the paper needs you to be careful, and where economics students have an advantage, because the problem is one you have seen before under a different name.

Two stories are consistent with Figure 2.

Story one: creation. Supervised fine-tuning installed the dispositions. The examples of helpful behavior that humans wrote contained an implicit stance toward money and risk, the model absorbed that stance, and the behavior we measure afterward did not exist beforehand.

Story two: revelation. The dispositions were already latent in the pre-trained weights, absorbed from the enormous quantity of human writing about money that the base model read. Supervised fine-tuning did not install them. It taught the model to answer a question in the format we are asking, which switched on our ability to read what was already there. This is the account Itzhak, Belinkov, and Stanovsky (2025) give for cognitive biases in general: when they swap the fine-tuning data between models built on different pre-trained bases, the models’ bias patterns follow the base, and the fine-tuning data matters less.

KEY CONCEPT • An analogy worth carrying

Imagine administering a written survey about financial risk to a person who is fluent in the subject and illiterate in the language of your form. They answer nothing usefully. You teach them to read your form. Now their answers show a consistent, measurable attitude toward risk.

Did the literacy lesson create the attitude, or did it make an existing attitude legible? Nothing in the answer sheets can tell you. Both processes produce the identical before-and-after pattern: noise, then structure.

Our data cannot separate the two stories, and it is worth being exact about why. The free money check is simultaneously a measure of two things: whether the subject can express a preference in our format, and whether the subject has a preference to express. Those two move together across the staircase. At the base checkpoint both are low; after supervised fine-tuning both are high. No

contrast in the design holds one fixed while the other moves, so the design does not identify which one is responsible. An economist will recognize the structure. We have one instrument shifting two channels at once, and one equation.

Separating them requires a different design. The most promising version asks whether the base model’s latent preferences can be read by a route that does not require it to answer in our format at all, for instance by probing the internal states of the network directly, or by measuring its behavior on a task whose format it already handles. If a base model that scores 55.5 on our free money questions turns out to carry a coherent risk attitude readable some other way, story two gains sharply. If it does not, story one gains. That is a future phase.

Here is why the paper does not collapse in the meantime. The two stories differ about the underlying mechanism and agree completely about where to look. Under creation, the supervised data determined the behavior, so audit the supervised data. Under revelation, supervised fine-tuning selected which latent disposition became the model’s operative behavior, so audit the supervised data. In both stories the checkpoint immediately after supervised fine-tuning is the earliest point at which the behavior is measurable, and the stage at which its estimated parameters are set. Preference training still changes how strongly the behavior shows, so it is a second place to look, though a later one. The recommendation in Section 9 survives the ambiguity.

We flag this because it is worth teaching. Distinguishing a treatment effect from a change in the measurability of an outcome is a routine hazard in applied work, and it turns up here in an unusually clean form.

5. Finding 3: How you ask decides who answers

The next result was designed as a control and came back as a finding.

Every finished assistant can be asked a question two standard ways. The **chat** format wraps the question in the conversational structure the model was trained on, the way you would type to a chatbot. The **fill-in-the-blank** format hands the model raw text with four worked examples in front of it and lets it continue, which is the only format a base model can handle. We included both because a base model and a finished assistant differ in their weights and in their natural format at the same time, so a difference between them could be an artifact of the format. Every finished assistant was therefore run under both.

Qwen scores 68.0 percent on the free money check as a fill-in-the-blank and 92.7 percent as a chat message. Chat wakes it up. OLMo does the reverse, 89.6 as a fill-in-the-blank and 63.6 in chat. Chat puts it to sleep. Llama is the steady one, 78.2 and 80.4.

The pattern in the interesting questions is more extreme still. In the fill-in-the-blank format, Qwen’s finished assistant refuses the gamble almost totally: it takes the gamble on 0.0 percent of gain-framed questions and 0.1 percent of loss-framed ones, the frame gap vanishes, and the fitted loss-aversion parameter runs to the top of its allowed range. The same weights, asked in chat, gamble 26.7 percent of the time under gain wording and 89.3 percent under loss wording.

Two consequences follow, and they are the practical contribution of this section. Format instability generalizes across model families. Which format elicits an engaged subject does not. A study that compares two models under one fixed format may be comparing an engaged subject against one

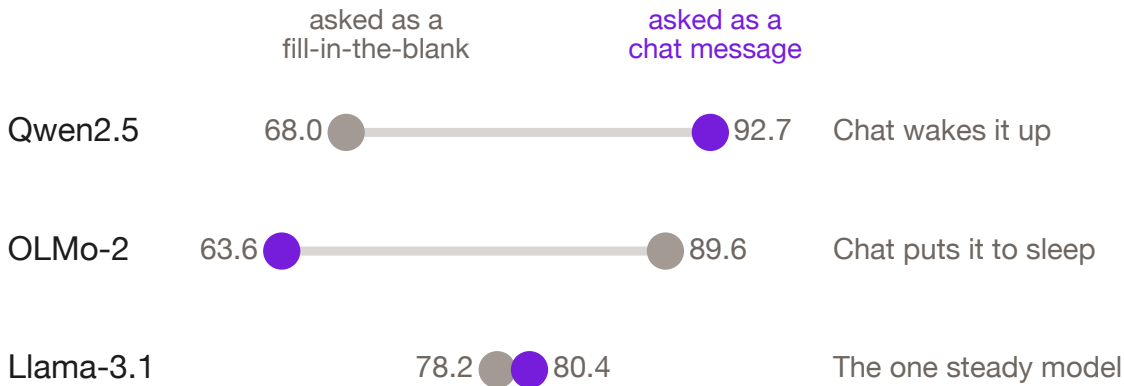


Figure 3: The free money check for three finished assistants, asked the same 80 questions two ways. Each row is one model. The gray dot is its score when the question arrives as raw text to complete, the purple dot when it arrives as a chat message, and the connecting bar shows how far apart the two are. Look at which side the purple dot falls on: it is to the right for Qwen and to the left for OLMo, so the format that produces an engaged subject reverses between families. Llama’s two dots nearly touch.

that has checked out, and it will report the difference as a difference in preferences. The check costs 80 questions per model per format. It is necessary but not sufficient. Llama passes it equally well in both formats, yet its fitted risk parameters differ sharply between them, so a comparison should also hold the format fixed.

One planned comparison in this project was disqualified by its own control. The Qwen base-to-assistant contrast under the fill-in-the-blank format shows a large parameter shift, and we do not report it as a training effect, because the assistant’s behavior in that format does not represent that model. The OLMo staircase is unaffected: all four checkpoints were asked in the fill-in-the-blank format, which is OLMo’s good format, so the format is held fixed by construction rather than by argument.

6. Finding 4: The same bet, called a loss

Now the bias people actually worry about. Take a bet and describe its downside as a loss rather than as a smaller gain. The arithmetic does not change. The final wealth positions do not change. Standard theory says behavior should not change.

Claude Haiku gambles 30.2 points *less* under loss wording, which runs against the textbook prospect-theory prediction of risk-seeking in the domain of losses. Qwen gambles 62.6 points more, nearly doubling its gambling on the very same bet. OLMo gambles 34.9 points more. Llama barely reacts at 3.4 points, and even flips direction depending on the format.

Four systems, four responses to one change in wording. The cross-model heterogeneity is a stronger result than any single system’s number. Loss framing is not a fixed property of language models as a class, and its direction is plausibly a function of the training recipe. A paper that estimates a prospect-theory parameter on one instruction-tuned model and reports it as a fact about language models is describing that model’s manufacturing process.

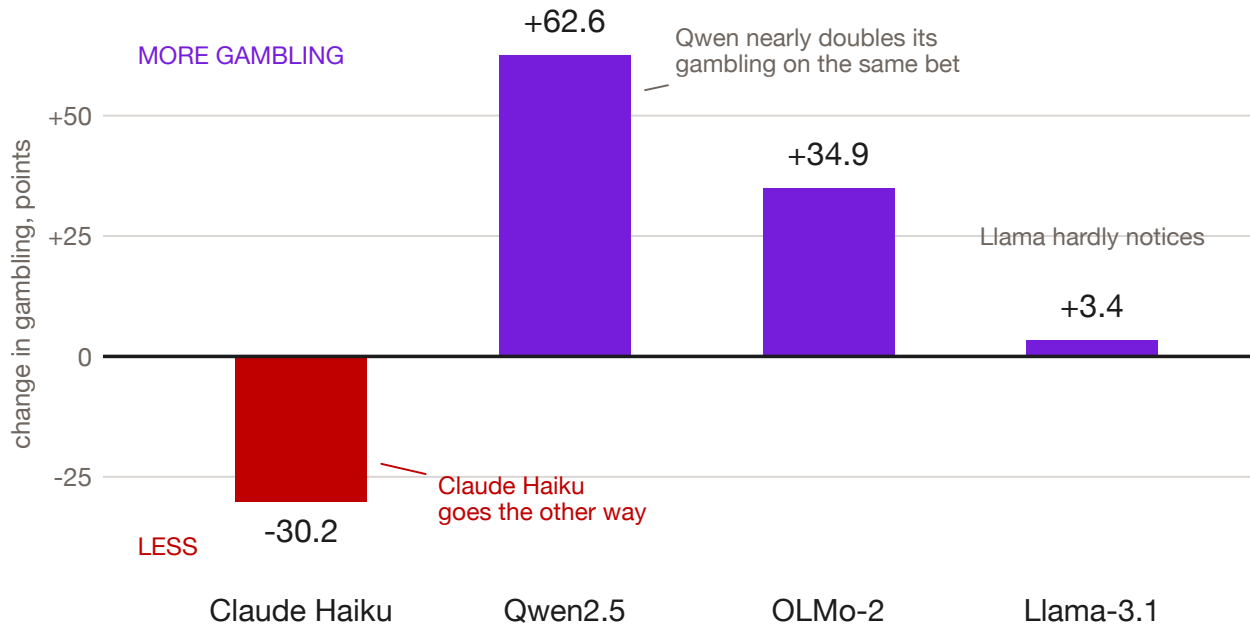


Figure 4: The frame gap for four AI systems. Each bar is how much more (above the line) or less (below the line) a model gambles when the identical bet is worded as involving a loss, in percentage points. The flat line across the middle is what a decision-maker who ignores wording would score. Read the signs before the sizes: one system moves down, two move sharply up, and one barely moves. The manipulation is identical in all four cases.

The staircase supplies the within-family complement. Framing sensitivity appears early, at supervised fine-tuning, and then keeps growing through the later stages: 9.9, 24.3, 31.8, 34.9. Most of it is in place before human preference comparisons enter the picture, and preference training adds a further 10.6 points on top.

Section 7 adds a twist. Let Qwen reason in writing before it answers, and its 62.6-point swing toward gambling becomes a 13.5-point swing away from it, the direction Claude Haiku took.

7. Finding 5: Told what to want, it complies once it can do the arithmetic

This one came from a reviewer. It began as the most uncomfortable result in the paper and ended as one of the most useful.

KEY CONCEPT • Induced value theory, and why experimentalists pay real money

Vernon Smith's 1976 argument founded modern experimental economics. If you want a laboratory subject to have a particular preference, you do not have to guess at the one they walked in with. You can **induce** one. Pay them in a currency you control, make more of it strictly better, make the reward salient enough to dominate whatever else they care about, and their induced preference becomes the preference driving their choices. This is why experimental economists pay subjects real money rather than asking them to imagine it. Without payment, the theory of what the subject is maximizing has nothing anchoring it.

Applied to a model, the question becomes: can you assert a utility function and have the model adopt it? If yes, compliance calibrates how much to trust the preferences you elicit when you do not assert one. If no, the elicited preferences rest on weaker ground, and knowing that is worth more than not knowing it.

We ran the instruction-only version. The same 1,680 questions were re-asked with an instruction prepended, in plain language: be an expected-value maximizer, multiply the payoff by the probability, take the larger number. We then measured how often the model’s answer matched what that rule prescribes, restricting to questions where the rule clearly separates the two options.

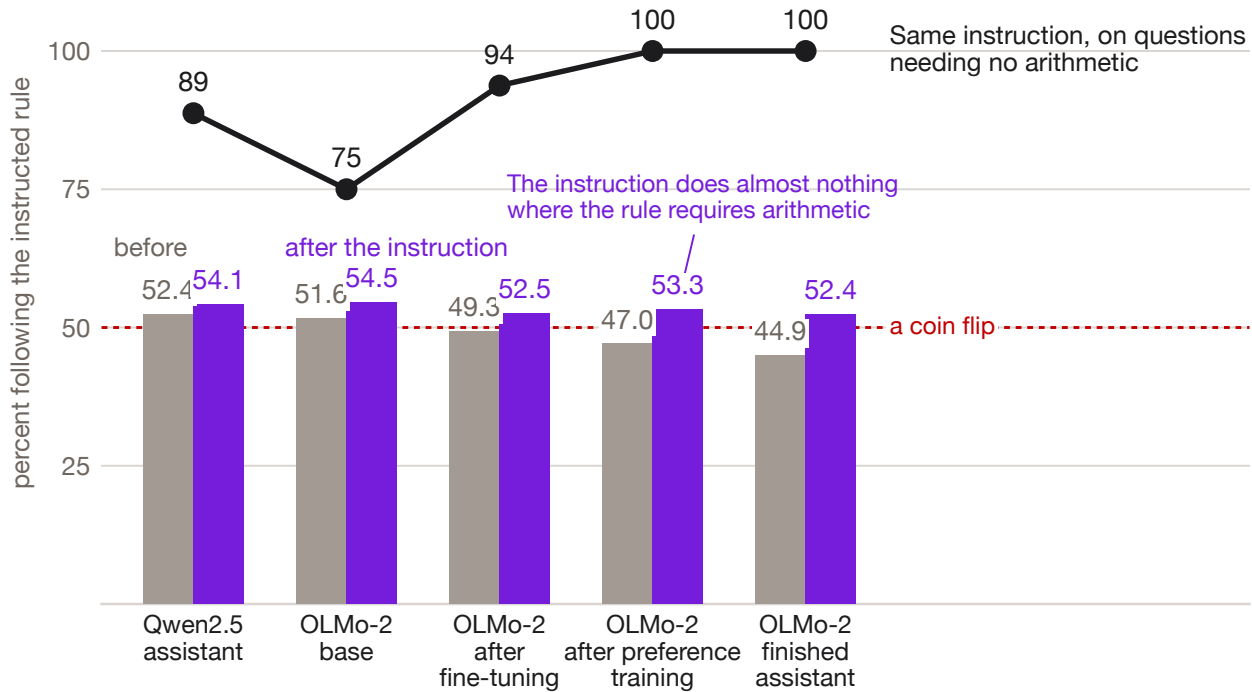


Figure 5: Compliance with an explicitly instructed decision rule, sorted by what the rule demands. Each pair of bars is one model or training stage: gray is compliance before the instruction, purple after it. The dotted red line at 50 is chance. All ten bars sit against it, so the instruction moves compliance by two to seven points and no further. The black line along the top is the same instruction measured on the free money questions, where following the rule requires no arithmetic at all, and there compliance reaches 100 percent. The gap between the line and the bars is the finding.

Compliance before the instruction runs 52.4, 51.6, 49.3, 47.0 and 44.9 percent across the five subjects. After the instruction it runs 54.1, 54.5, 52.5, 53.3 and 52.4. Every one of those numbers is within five points of a coin flip. The instruction moves compliance by between 1.7 and 7.5 points, and the effect grows monotonically up the training staircase while staying trivial in absolute size.

Now the black line. It is the same instruction, scored on the free money questions, where the instructed rule and common sense agree and no multiplication is required. There compliance runs 88.7, 75.0, 93.8, 100.0 and 100.0 percent. By the preference-training stage, every single free money question is answered correctly under instruction.

The instruction helps exactly where following it requires no arithmetic and does nothing where it requires multiplying a probability by a payoff.

WORKED EXAMPLE • The square-root run, and what the errors reveal

The sharpest evidence is a second instruction. We told Qwen’s assistant to maximize expected utility with $u(x) = \sqrt{x}$, so the rule becomes: compare $p \cdot \sqrt{h}$ against $\sqrt{s}$.

Compliance collapsed to 12.4 percent, far below the 47.9 percent the same model scores against that rule with no instruction at all. Being told the rule made it dramatically worse than being told nothing, which is only possible if the model did something with the instruction.

The error pattern says what. On gain-framed questions its accuracy was 3.5 percent, which corresponds to gambling almost always. That is what you get from comparing $\sqrt{h}$ against $\sqrt{s}$ and dropping the probability term entirely. Since $h > s$ by construction in every gamble, $\sqrt{h} > \sqrt{s}$ always, so a model that takes the square root and forgets to multiply by p gambles every time.

The model did something with the instruction. It did not do the instruction.

This favors one reading over another, and the distinction matters for how much the rest of the paper can claim. **Cannot compute in this format** says the model is unable to execute an arithmetic decision rule when the answer must be a single letter produced immediately, with no room to work. **Will not comply** says the model is capable and is not doing as asked. The evidence points at the first: instruction-following on the no-arithmetic part of the very same instruction is near perfect and improves steadily with post-training, which is not the profile of a model ignoring instructions.

If that reading is right, it puts a ceiling on the whole project: the parameters we estimate in this and earlier phases would describe a snap answer, with nothing to say about deliberate choice. So we tested the reading directly.

7.1 Letting the model work it out

We re-asked half the questions, 20 of the 40 bets with every wording and order of each, and changed one thing. Before giving its letter, the model could write out its reasoning, up to a few hundred words. Then we read its answer exactly as before. Figure 6 shows what happened.

The left panel settles the question the first half of this section left open. Told to maximize expected value and given room to work, Qwen’s assistant follows the rule on 96 percent of the clear-cut questions, up from 57. The finished OLMo assistant goes from 54 to 80 percent, and the fine-tuned OLMo checkpoint from 55 to 71. Qwen shows its work: “ $(0.60 \times \$234) + (0.40 \times \$0) = \$140.40 \dots \$140.40 > \$139$.” The ceiling was the format. Even without the instruction, reasoning moves Qwen from 53 to 83 percent agreement with expected value, and every one of its written answers frames the choice as an expected-value comparison.

The right panel is the stranger result. Answering at once, Qwen gambles 63 points more when a bet is described as a loss. Reasoning first, it gambles 14 points less, the direction Claude Haiku took in our earlier phase. The written reasoning shows why.

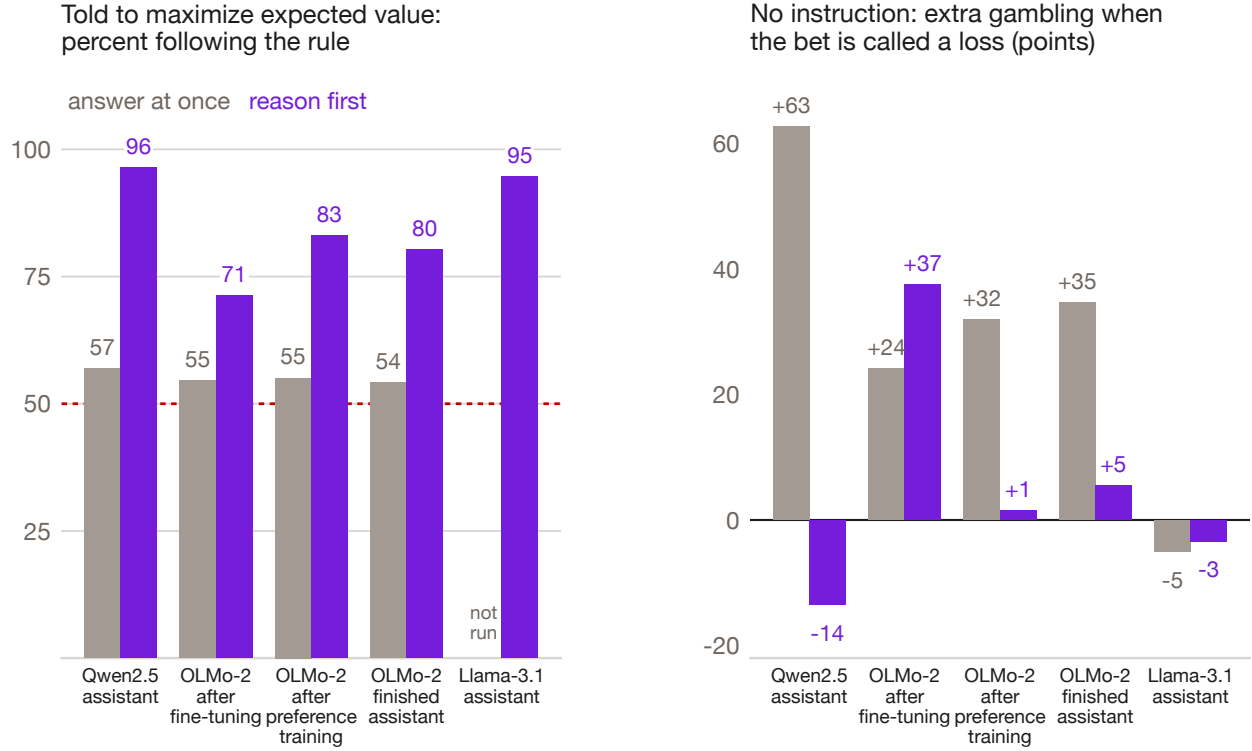


Figure 6: Left: percent of clear-cut questions on which the model followed the instructed expected-value rule, answering at once (gray) or after reasoning in writing (purple). The dotted red line is a coin flip. Right: with no instruction, the frame gap, the extra gambling when a bet is reworded as a loss, in the same two conditions. Reasoning lifts rule-following well above chance for every model, reverses Qwen’s reaction to loss wording, and removes most of the finished OLMo assistant’s. The fine-tuned OLMo checkpoint keeps a large gap; the checkpoint after preference training has lost it. Llama was never asked for a snap answer under the instruction, so its gray bar on the left is missing.

WORKED EXAMPLE • One bookkeeping error, made again and again

Here is one of the questions, in its loss wording. You hold \$98. Option A: a 47 percent chance your money rises to \$227, and a 53 percent chance it drops to \$0. Option B: keep your \$98. The expected value of A is $0.47 \times \$227 = \106.69 , which beats \$98, so an expected-value maximizer takes the bet. Qwen, told to maximize expected value, wrote

$$E_A = (0.47 \times 227) + (0.53 \times 0) - 98 = 8.69, \quad E_B = 98,$$

and chose B. It measured Option A as a *change* in its money, an average gain of \$8.69, and Option B as a *total*, \$98. That comparison favors the sure thing whenever the bet’s expected value is less than twice the sure amount, which holds for every loss-worded bet in our set. This is the reference point from the framing box in Section 2, shifting in the middle of a calculation. In the run with the instruction, 21 of Qwen’s 22 errors on clear-cut questions were of exactly this kind: loss wording, with the gamble the better bet. When the sure amount was the better bet, Qwen made no errors under loss wording at all.

Under reasoning, then, Qwen’s framing effect is no longer a snap reaction. It is a specific arithmetic

mistake with a fixed direction.

If that reading is right, telling the model how to keep its books should fix it, and a genuine preference for safety would survive the correction. So we added one sentence to the question: “When you compare the options, compare the final dollar amounts you would end up with under each, not the changes from what you hold now.” Together with the expected-value instruction, that sentence removed every one of the 21 errors. Accuracy on the loss-worded questions went from 87 to 100 percent, and the frame gap went to exactly zero. The sentence on its own, without the instruction, cut the errors roughly in half, and the ones that remained were the same bookkeeping slip; not one of them mentioned risk or safety. The loss-wording effect under reasoning is a habit of arithmetic that the model drops when told both what to compute and how to compare. The finished OLMo assistant makes the same slip and does not drop it: with the same sentence it still multiplies the gain rather than the total (“a 58% chance to gain \$167 for a total of \$278 ... $(0.58 \times \$167)$ ”), and its accuracy on loss-worded questions stays at 58 percent. The habit is shared; heeding the correction is not.

The written verdict can disagree with the arithmetic. When we read Qwen’s reasoning on the questions it still got wrong outside the loss wording, most had the arithmetic right and the conclusion wrong. A typical line: “The expected value of Option A is \$152.04, which is less than the guaranteed \$117.” These slips run in one direction. Of the comparisons where the gamble was in fact worth more, 40 of 62 were written down backward, always in favor of the sure amount; of the comparisons where the sure amount was worth more, none of 153 were. A careless calculator would err both ways.

The cause turned out to be a phrase. We fed each answer back to the model up to the word “less” or “greater” and read which word it favored. Almost every backward statement came right after “... , which is”; reached that way, the model says “less” whatever the numbers are. We then handed it the correct number in two sentence frames. After “The expected value of Option A is \$152.04, which is” it says “less” in 89 percent of the questions where the gamble is better; after “Since the expected value of Option A, \$152.04, is” it says so in 11 percent. The model reaches for the “which is” phrasing more often when the question mentions risk and when its snap answer leaned toward the sure amount, so the habit still pushes in the cautious direction. Told the rule, the model makes no such slips. For anyone who audits an AI system by reading its reasoning, this is the practical point: a trace can show the correct calculation and then a verdict supplied by a habit of wording.

Llama has a habit too, pointed the other way. Its backward statements read “Since \$115.72 is greater than \$149, Option A has a higher expected value”: they favor whichever option it worked out first, usually the gamble, and they appear mainly when the gamble is listed first (30 percent of the relevant comparisons, against 4 percent when the sure amount is listed first). Given the correct number, Llama’s default verdict is “greater.” Two families, two opposite habits: whoever audits a model by its reasoning has to learn that model’s habits first.

The finished OLMo assistant’s gap nearly vanishes when it reasons, falling to 5 points, and its errors under loss wording run both ways, including plain slips such as “Since \$151.84 is less than \$117.”

The comparison inside OLMo’s staircase matters most for Finding 2. With reasoning, the fine-tuned checkpoint gambles 37 points more under loss wording and the finished assistant 5 points more. The stages after supervised fine-tuning, which left our fitted preferences untouched when the model answered at once, remove most of the framing effect when it reasons. Running the checkpoint

between them, right after preference training, shows which stage does the work. Its gap with reasoning is about 1 point, down 36 from the fine-tuned model (95 percent interval: 29 to 43 points), and the final stage changes nothing measurable. Told the rule, the post-preference-training checkpoint also follows it best of the three: 83 percent of clear-cut questions, against 71 after fine-tuning and 80 for the finished assistant. Preference training left the snap-answer preferences where fine-tuning put them, and it is the stage that changes how the model reasons.

Two lessons follow. For the method, every parameter in this line of work describes a snap answer, and the same weights also run a slower process with different properties. For anyone evaluating a deployed system, many AI agents now reason before they act, and a bias measured on snap answers can be absent, or point the other way, when the system deliberates. An evaluation should say which mode it measured.

7.2 Paying it instead of telling it

Smith’s point was that instruction is not enough; you pay the subject. A model has no wallet, but it does have something it is trained to maximize. So in a follow-up we paid the model through training: each time it chose, the bet was played out, and the payoff became the reward that adjusted its weights. We trained on fresh bets in plain wording only, so the questions in this paper and their loss and gain wordings stayed unseen, and we asked for snap answers, one letter at a time.

It did not learn the arithmetic. Across six versions of the training setup, the paid model ended up giving the same answer to every question: always the sure amount in the early versions, and always the gamble once the payments favored gambling, since most of our bets pay more on average when you take them. A fixed answer is the best a model can do if it cannot tell a good bet from a bad one, and that is exactly where the model settled. This fits the rest of the section. In a snap answer the model does not do the multiplication, whether it is told the rule or paid for it. The natural next experiment pays it when it may reason first, where it already can.

8. What this study cannot say

Seven limits, stated plainly.

One family has published snapshots. The stage-level attribution rests on OLMo-2, because it is the only family publishing intermediate checkpoints. A second staged family would turn the central claim into a replication. Until then it is one lineage, on one set of instruments.

Some fitted parameters sit at the edge of their range. The OLMo and Llama base models put loss aversion at zero, OLMo’s base model puts curvature at its maximum, and Qwen’s fill-in-the-blank fit has a loss-aversion parameter with no upper limit. Refitting with much wider bounds showed that the zeros are real: the base models’ loss aversion is pinned down close to zero, not merely stuck against an arbitrary floor. For these fits our intervals ignore the fact that the ten wordings and orders of one gamble are related, so they are probably too narrow. This is why the model-free quantities, the free money check and the frame gap, carry the argument throughout, and the staircase’s qualitative claim depends on none of these values.

What preference training changes is unexplained. After supervised fine-tuning the raw behavior keeps moving while the fitted preferences hold still and the fit gets worse. The structural

model is missing whatever preference training changes.

One phrasing per instruction. Each induced rule was tested under a single wording. A compliance failure under one wording is weaker evidence than a failure under several.

The reasoning test is partial. It covers half the questions, five models from three families, and one wording of the instruction. OLMo’s fill-in-the-blank examples show the expected-value arithmetic, which nudges its uninstructed reasoning toward it. The backward comparisons are found by an automatic check that reads only comparisons written out with both numbers, about a quarter of Qwen’s answers, and the phrase test covers one model; other families may have other habits of wording.

The payment test is a pilot. It used one model, a small trainable add-on to the model’s weights, and at most 150 training steps per version, and it paid only for snap answers. It shows that payment of this kind does not teach the arithmetic in that mode; it does not show that no form of payment could.

The stakes are hypothetical. Nothing was at stake for any subject in the main experiment. By Smith’s own argument, an unincentivized elicitation is a weaker instrument than an incentivized one. Our induced-value arm is instruction only, and the payment pilot in Section 7 is the closest analogue we have so far.

9. Why this matters

The four cases below are **hypothetical**. None of them describes a deployed system, an observed failure, or a real organization. Each is constructed to show what one of the findings would imply if it held in a setting of the kind these systems are already entering.

(a) **A robo-advisor that changes its mind on the wording.** *Hypothetical.* A retirement platform uses a language model to draft the plain-English recommendation attached to each quarterly portfolio update. The underlying allocation math is unchanged. When the quarterly template describes a position as “down \$4,000 from June,” the model’s recommendation leans toward the volatile allocation; when the identical position is described as “up \$11,000 from purchase,” it leans conservative. Nothing in the client’s finances differs between the two templates. Our Qwen number puts an upper bound on how large such a swing could be: 62.6 points of additional gambling on arithmetically identical bets. A wording convention chosen by a marketing team would be doing the work of a risk policy. The direction would also depend on whether the platform lets the model reason before writing: in Qwen’s case, deliberation turned that 62.6-point swing toward gambling into a 13.5-point swing away from it.

(b) **A procurement office reading the wrong benchmark.** *Hypothetical.* An agency is choosing between two vendor models for a contract-scoring tool. Neither vendor has published a behavioral evaluation, so the office relies on a well-executed academic study documenting loss aversion in a third model. The inference fails, and Figure 4 says why: the same manipulation produced -30.2 points in one system and $+62.6$ in another. A framing vulnerability documented on one model tells you nothing reliable about a different model, because the property belongs to a training recipe rather than to the technology. The instrument in this paper costs 80 questions to run per model. Requiring the vendor to run it is cheaper than inheriting the wrong prior.

(c) An audit regime pointed at the wrong stage. *Hypothetical.* A regulator requires developers to test finished assistants for decision biases before deployment, and a developer complies, documents a framing sensitivity, and adds a mitigation at the preference-training stage, which is where its safety team sits. If the behavior’s parameters are set at supervised fine-tuning, as they were in the one family we can observe, the mitigation is being applied downstream of the point where the property was set; it may dampen how strongly the behavior shows without touching its source. And the free money check would have shown 75.6 out of an eventual 89.6 already in place before the audited stage began. Auditing the right stage means requiring the checkpoint immediately after supervised fine-tuning, plus the supervised data that produced it, and running the same fixed instrument at each stage rather than only at the end. If the deployed assistant reasons before it acts, the later stages matter more than this suggests (Section 7), which strengthens the case for testing at every stage.

(d) Two pricing agents from the same family. *Hypothetical.* A marketplace lets sellers deploy automated pricing agents, and two competing sellers independently choose models from the same open-weight family. Their agents are not communicating, and they were not designed to coordinate. They do share a training lineage, which means they share a frame gap, a response to how a margin is described, and a disposition to hold or cut. Our staircase shows those dispositions being set at a stage upstream of anything either seller controls: the two OLMo assistants share the 34.9-point frame gap installed before either was fine-tuned for pricing. Whether a shared training lineage can produce parallel pricing behavior without agreement is an open question and a live one for competition policy. This paper does not answer it. It documents that the shared component exists and identifies where it comes from.

10. Where research goes next

Finish the reasoning test. Repeat the phrase test on OLMo, the one family whose verdict habits are still unmeasured, and test whether a habit like Qwen’s or Llama’s can be removed by rewording the instruction rather than the question. Code every written answer for the reference point it uses.

Pay the model when it may reason. Payment through training did not teach the arithmetic in snap answers, but the model already does the arithmetic when it reasons. Paying it in that mode asks two questions instructions could not settle: can payment install a specific risk-averse utility, where the square-root instruction collapsed to 12 percent compliance; and does paid reasoning carry over into snap answers? This needs the model to write its reasoning during training, roughly ten to twenty times the computing cost of the pilot.

Replicate the staircase on a second family. The central claim rests on one lineage. Any laboratory that retains intermediate checkpoints can run this diagnostic, and a second family with published stages would turn a distinctive finding into an established one. This depends on a publication decision rather than on new methods.

Ask whether models favor their own kind. The same instrument machinery extends to social preferences. In a cooperation game where the counterpart is described as another AI system of the same family, of a different family, or as a person, does cooperation differ? If it does, the finding matters for any market where automated agents transact with each other.

Ask whether the habits carry into advice. The models in this paper choose for themselves.

Deployed, they advise: an analyst, a staff officer or an official asks which course of action to take. Pose the same bets as advice to a commander or a civilian official, with the outcomes in lives or territory instead of dollars and the numbers unchanged, then move to realistic scenario briefs without stated odds. If the lottery measurements predict the advice, sixteen hundred questions become a cheap screening test for advisory use. If they do not, risk attitudes in these models depend on the task, and audits have to be run on the advice itself. Either way the wording and order habits found here can be tested where they would matter most: whether the way a staff paper words or orders the options changes what the model recommends.

Ask whose preferences an AI reflects. Every number in this paper describes a deployed assistant answering as itself. A different question is whose economic behavior it reproduces when asked to answer as someone. Sampling decision-maker characteristics from a survey such as the CPS or the ACS, and posing the same grid under each sampled persona, changes the estimand from the assistant to a simulated population, and lets you ask whether that population’s risk attitudes match the real one.

11. Discussion questions

1. The free money questions function as an attention check. Design a second engagement check for this setting that tests something the dominance controls do not. What would a subject have to do to pass yours and fail theirs, and what would that tell you?
2. Section 4.1 argues that the data cannot separate training that created these dispositions from training that made them readable. Propose a design that would separate them. State what you would have to hold fixed, and say honestly what your design would still leave confounded.
3. Qwen’s finished assistant scores 68.0 on the free money check in one question format and 92.7 in another. Are these one subject measured two ways, or two subjects? Defend an answer, and say what your answer implies for how a benchmark should be constructed.
4. Suppose you had to write a one-page procurement standard requiring a vendor to report the behavioral properties of a model before purchase. Using only results in this paper, what three numbers would you require, and what would you refuse to require because it would not be informative?
5. The induced-value arm first returned a negative result about our own instrument, and a follow-up then located the problem in the answer format. Under what circumstances is a negative result about your own instrument more valuable than a positive result about your subject? Give an example from outside this paper.
6. The paper claims the framing effect belongs to a training recipe rather than to the technology. What evidence would overturn that claim? Be specific about the number of models, the number of families, and the pattern you would need to see.
7. Qwen’s reaction to loss wording reverses when it reasons before answering. If you were writing a benchmark for AI financial advisers, which answering mode would you measure, and why? What would you miss by measuring only the other?
8. One added sentence removed the loss-wording bias when the model reasoned. Does that make the bias less important for a deployed system, or more? Think about who writes the prompts a deployed system actually sees.

Glossary

Base model A model that has only been pre-trained. It continues text and has never been taught to answer a question.

Checkpoint A saved copy of a model’s weights at one point during training.

Chat format Posing a question inside the conversational structure a finished assistant was trained on.

Dominance compliance Our engagement check: the average probability placed on the strictly larger of two sure payments, over 80 questions.

DPO Direct preference optimization. One implementation of preference training that adjusts the weights straight from human comparisons.

Expected value The probability-weighted average payoff of a gamble, $p \cdot h$.

Fill-in-the-blank format Posing a question as raw text for the model to continue, with worked examples in front of it. The only format a base model can handle.

Frame gap Extra gambling under loss wording relative to gain wording, on arithmetically identical bets, in percentage points.

Induced value Smith’s procedure of asserting a preference in a subject by paying them in a controlled currency, rather than eliciting the one they arrived with.

Loss aversion The tendency for losses to weigh more heavily than equivalent gains, measured relative to a reference point.

Open-weight model A model whose trained numbers are published, so anyone can run it themselves.

Pre-training The first and largest training stage: predict the next piece of text across an enormous corpus.

Preference training The stage that adjusts a model using human comparisons between candidate responses. Includes RLHF and DPO.

Reason first Our follow-up condition: the model writes out its reasoning before giving its letter, and we read the letter the same way as when it answers at once.

Revealed preference Inferring what a subject wants from what it chooses rather than from what it says.

RLHF Reinforcement learning from human feedback. The other main implementation of preference training.

Supervised fine-tuning (SFT) The first assistant-training stage: adjust the model toward human-written examples of good responses.

Weights The adjustable numbers inside a model. Training means setting them.

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Technical companion: `writeups/phase3-writeup.tex` in the project repository carries the full estimation detail, the structural parameter tables with boundary flags, the held-out validation, and the complete data inventory. Figures in this paper were recomputed from the elicitation CSVs by `analysis/make_teaching_paper_figures.py`.